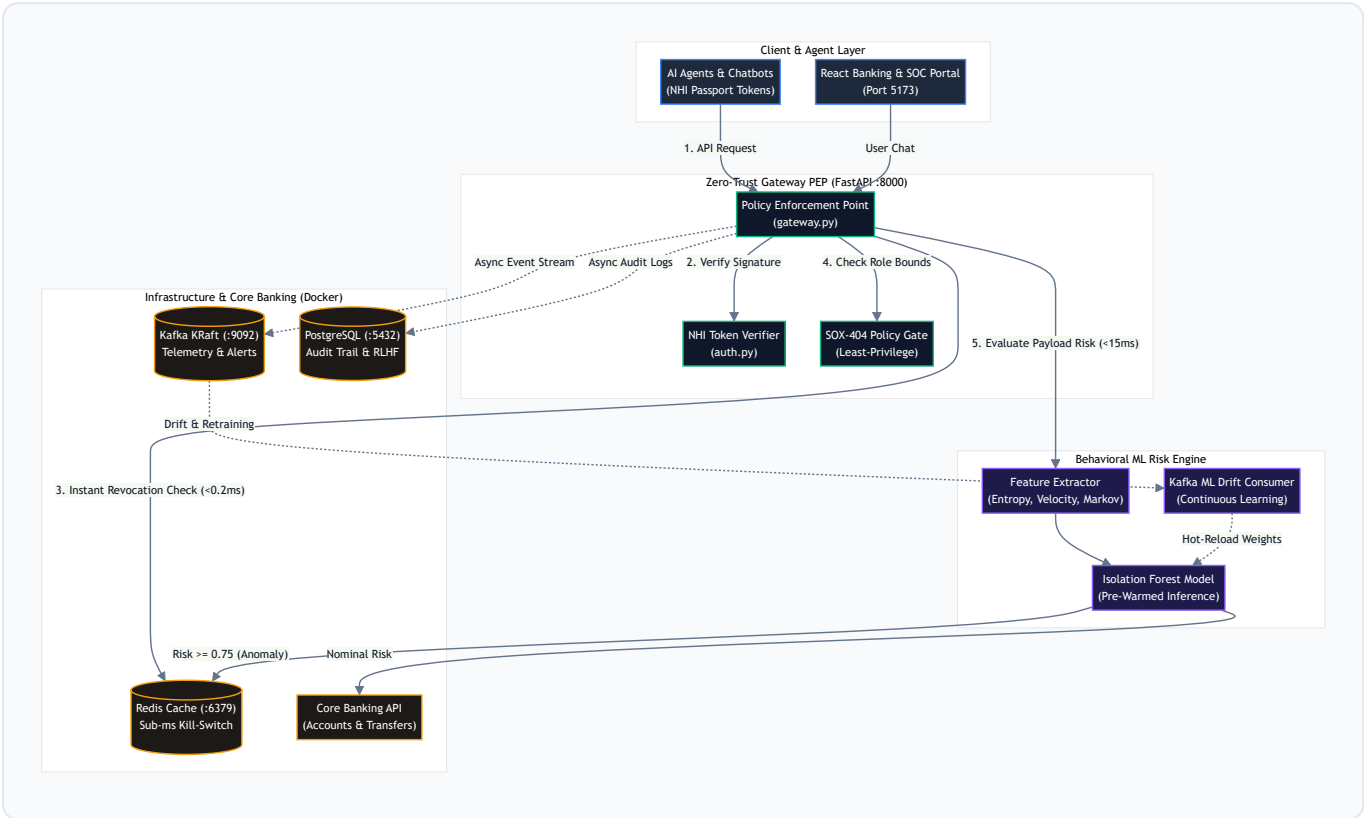


Zero-Trust Non-Human Identity (NHI) Access Governor

BFSI Autonomous AI Agent Security & Policy Enforcement Point

Deloitte Capstone Project • High-Level Presentation Architecture • September 2026

1. High-Level System Architecture Diagram



2. Four Core Architectural Pillars

Pillar 1: Agent Identity & Tokenization

Cryptographically signed HMAC-SHA256 Non-Human Identity (NHI) Passports with role claims (tier1_customer_service, payment_operations) and transaction limits.

Pillar 2: High-Performance Gateway PEP

FastAPI reverse proxy serving fast-path GET queries in <2.0ms. Probes Redis revocation cache in <0.2ms and enforces deterministic banking least-privilege.

Pillar 3: Behavioral ML Risk Engine

Pre-warmed Isolation Forest model evaluating Shannon entropy, request velocity bursts, and Markov API state jumps in 12–15ms.

Pillar 4: Event-Driven Infrastructure (Docker)

PostgreSQL for compliance audit logs and RLHF conversations, Redis for kill-switch blacklisting, and Kafka KRaft streaming telemetry to the ML Drift Consumer.

SOX-404 Compliance Highlight: Enforces Segregation of Duties (SoD) directly in the Policy Enforcement Point. Support bots are deterministically blocked from moving funds or liquidating assets with zero external RPC overhead.

3. Production Benchmarks & Verified SLAs

Governance Metric	Target / SLA	Verified Production Benchmark	Compliance Standard
Nominal PEP Latency (GET)	< 5.0 ms	< 2.0 ms	Deloitte Fast-Path SLA
Full ML Risk Inference (POST)	< 30.0 ms	12.0 – 15.0 ms	Behavioral Anomaly Detection
Kill-Switch Remediation (MTTR)	< 200.0 ms	< 5.0 ms	FFIEC Rapid Quarantine
Revocation Cache Probe	< 1.0 ms	< 0.2 ms	Redis Key Blacklist
Automated Test Suite	100% Pass	58 / 58 Passed (0 warnings)	Continuous Integration